Ans. (a) : IRNSS is an independent regional navigation satellite system being developed by India. It is designed to provide accurate position information service to the user in India as well as the region extending upto 1500 km from its boundary, which is its primary service area. Statement (2) is incorrect.

- Out of 8 satellites, 3 satellites are located in suitable orbital slots in the geostationary orbit and the remaining 4 are located in geosynchronous orbit, with the required inclination and equatorial crossings in two different planes- So, statement (1) is correct.
- There are plans to add more satellites to the IRNSS and expand it to the SAARC region by 2019. So statement (3) is incorrect.

76. Consider the following pairs :

	Terms sometimes seen in news		**Context/Topic**
1.	Belle II experiment	–	Artificial Intelligence
2.	Blockchain technology	–	Digital/Cryptocurrency
3.	CRISPR-Cas9	–	Particle Physics

Which of the pairs given above is/are correctly matched ?

(a) 1 and 3 only (b) 2 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans. (b): The Belle II experiment is a particle physics experiment designed to study the properties of B mesons (heavy particles containing a bottom quark). It is currently being commissioned at the Super KEKB accelerator complex at KEK in Tsukuba, Ibaraki, Prefecture, Japan.

Blockchain technology can be integrated into multiple areas. The primary use of blockchains today is as a distributed ledger for cryptocurrencies, most notably bitcoins.

- CRISPR- Cas 9 is a genome- editing tool. It is short for Clustered regularly interspaced short palindromic repeats and CRISPR-associated protein.

77. Which of the following statements best describes "carbon fertilization?

(a) Increased plant growth due to increased concentration of carbon dioxide in the atmosphere
(b) Increased temperature of Earth due to increased concentration of carbon dioxide in the atmosphere
(c) Increased acidity of oceans as a result of increased concentration of carbon dioxide in the atmosphere
(d) Adaptation of all living beings on Earth to the climate change brought about by the increased concentration of carbon dioxide in the atmosphere

Ans. (a): The carbon fertilization effects suggests that the increase of carbon dioxide in the atmosphere increases the rate of photosynthesis in plants. The effect varies depending on the plant species, the temperature and the availability of water and nutrients.
A quarter to half of earth's vegetated lands has shown significant greening over the last 35 years, largely due to the rising level of atmospheric carbon dioxide.

78. When the alarm of your smart phone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a message to your office accordingly.
In the context of emerging communication technologies, which one of the following terms best applies to the above scenario?

(a) Border Gateway Protocol
(b) Internet of Things
(c) Internet Protocol
(d) Virtual Private Network

Ans. (b): The Internet of Things (IOT) is the network of physical devices, vehicles, home appliances and other items embedded with electronics, software, sensors and connectivity. They enable these devices to connect and exchange data creating opportunities for more direct integration of the physical world into computer-based systems resulting in efficiency improvements, economic benefits and reduced human intervention.
Often, these devices use Internet Protocol (IP) the same protocol that identifies computers over the world wide web and allows them to communicate with one another.

79. With reference to solar power production in India, consider the following statements :

1. **India is the third largest in the world in the manufacture of silicon wafers used in photovoltaic units.**
2. **The solar power tariffs are determined by the Solar Energy Corporation of India.**

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d): Not a single silicon chip for solar energy is manufactured in India. Every solar panel which is made in India is assembled, while all the material comes from China, Europe and some other countries. India is yet to develop semiconductor clusters. Statement (1) is incorrect.

- CERC determines solar power tariffs and regulates the tariff of generating companies owned or controlled by the central Government.
- SECI has a power trading license, but it does not set solar power tariffs.

80. "3D printing" has applications in which of the following?

1. **Preparation of confectionery items**
2. **Manufacture of bionic ears**
3. **Automotive industry**
4. **Reconstructive surgeries**
5. **Data processing technologies**

Select the correct answer using the code given below :

(a) 1, 3 and 4 only (b) 2, 3 and 5 only
(c) 1 and 4 only (d) 1, 2, 3, 4 and 5

Ans. (d) : 3D printing is being used in a wide range of industries, from construction to medical technology.
Some of the applications of 3D printing:

- Through 3D printing, we can create ornamental prototypes, starting from construction drawings, as well as mechanical plants.
- 3D printing of confectionery has focused primarily on chocolate and hard sugar candy, and recent innovations have led to the printing of gummy candy.
- The 3D printed bionic ear that can "hear" radio waves.
- 3 D printing is Facial Reconstructive Surgery, people can now receive custom 3D-printed facial implants to help after illness or injuries.
- Multilateral voxel-printing method enabling the physical visualization of data sets commonly associated with scientific imaging.
- It is also used in Maritime Industry, Electronics and Entertainment and Broadcasting etc.

81. With reference to India's satellite launch vehicles, consider the following statements :

1. **PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.**
2. **Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.**
3. **GSLV Mk III is a four-staged launch vehicle with the first and third stages using solid rocket motors; and the second and fourth stages using liquid rocket engines.**

Which of the statements given above is/are correct ?

(a) 1 only (b) 2 and 3
(c) 1 and 2 (d) 3 only

Ans. (a) : PSLV is designed mainly to deliver the "earth-observation" or "remote-sensing" satellites with a lift-off mass of up to about 1750 kg to Sun-Synchronous circular polar orbits of 600-900 km altitude. The GSLV is designed mainly to deliver the communication-satellites to the highly elliptical (typically 250×3,6000 km) Geosynchronous Transfer Orbit (GTO). The satellite in GTO is further raised to its final destination viz, Geo-synchronous Earth Orbit (GEO) of about 3,6000 km altitude (and zero degrees inclination on the equatorial plane) by firing its in-built on-board engines. Due to their geo-synchronous nature the satellites in these orbits appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth, thus avoiding the need for a tracking ground antenna and hence are useful for communication applications.
GSLV MK-III is a three-stage heavy- lift launch vehicle developed by ISRO. The vehicle has two solid strap-one, a core liquid booster and a cryogenic upper stage.

ENVIRONMENT AND ECOLOGY

82. How is the National Green Tribunal (NGT) different from the Central Pollution Control Board (CPCB)?

1. **The NGT has been established by an Act where as the CPCB has been created by an executive order of the Government.**
2. **The NGT provides environmental justice and helps reduce the burden of litigation in the higher courts whereas the CPCB promotes cleanliness of streams and wells, and aims to improve the quality of air in the country.**

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (b): The National Green Tribunal (NGT) has been established on 18.10.2010 under the National Green Tribunal Act 2010 for effective and expeditious disposal of cases relating to environmental protection and conservation of forests and other natural resources, including enforcement of any legal right relating to the environment and giving relief and compensation for damages to persons and property and for matters connected therewith or incidental thereto.
The tribunal's dedicated jurisdiction in environmental matters shall provide speedy environmental justice and help reduce the burden of litigation in the high courts. The tribunal is mandated to make and endeavour for disposal or applications or appeals finally within 6 months of filing of the same.
The CPCB statutory organisation was constituted in September, 1974 under the Water (Prevention and Control of Pollution) Act, 1974. Further, CPCB was entrusted with the powers and functions under the Air (Prevention and Control of Pollution) Act, 1981.
Functions of the CPCB, as spelt out in the water (Prevention and Control of Pollution) Act,1974, and the Air (Prevention and Control of Pollution) Act, 1981 (i) to promote cleanliness of streams and wells in different areas of the states by prevention, control and abatement of water pollution and (ii) to improve the quality of air and to prevent, control as abate air pollution in the country. Hence, statement (1) is incorrect and statement (2) is correct.

83. Consider the following statements:

1. **Most of the world's coral reefs are in tropical waters.**
2. **More than one-third of the world's coral reefs are located in the territories of Australia, Indonesia and Philippines.**
3. **Coral reefs host far more number of animal phyla than those hosted by tropical rainforests.**

Which of the statements given above is/are correct?

(a) 1 and 2 only (b) 3 only
(c) 1 and 3 only (d) 1, 2 and 3